VLSI Devices
Lecture 15

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L15

Effective mobility

- Previously, we made the following simplification:

$$I_d(y) = qW\int_0^{x_i} \mu_n n(x,y)\frac{dV}{dy}dx = -\mu_{eff}W\frac{dV}{dy}\left(-q\int_0^{x_i} n(x,y)dx\right)$$

$$= -\mu_{eff}W\frac{dV}{dy}Q_i(y)$$

Taur, Eq. (5.8)

  - Then, what is μ_{eff}?

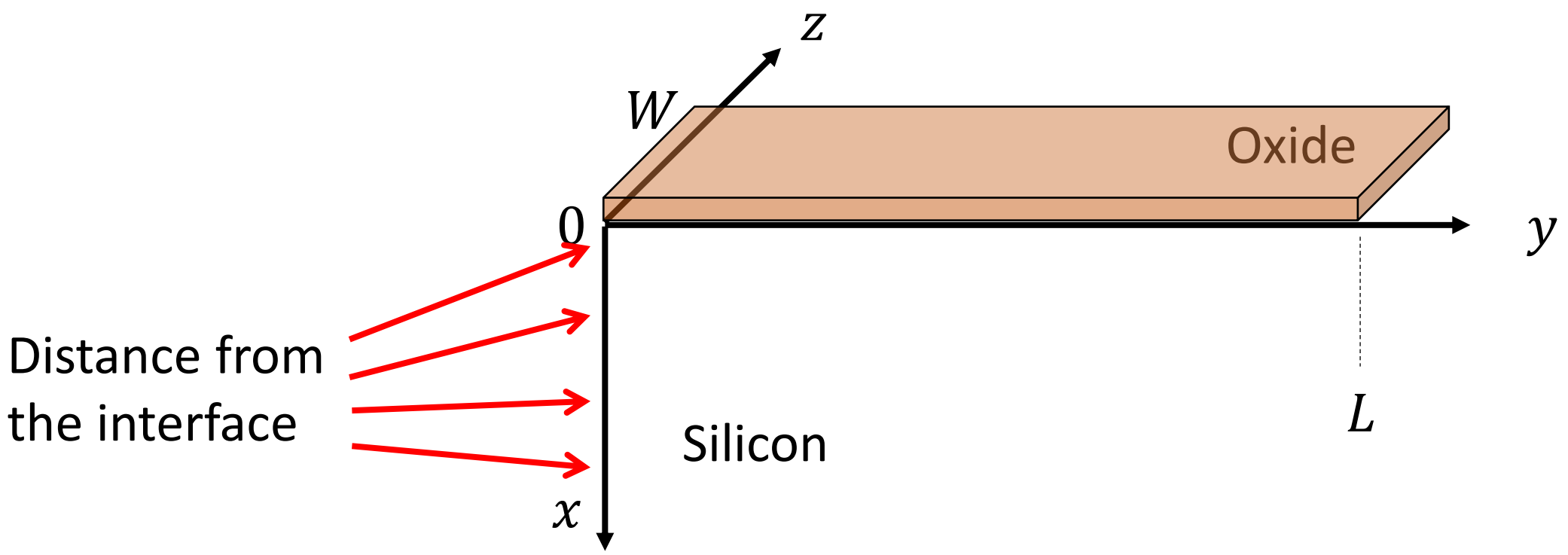
$$\mu_{eff}(y) = \frac{\int_0^{x_i} \mu_n n(x,y)dx}{\int_0^{x_i} n(x,y)dx}$$

Taur, Eq. (5.50)

Position-dependent

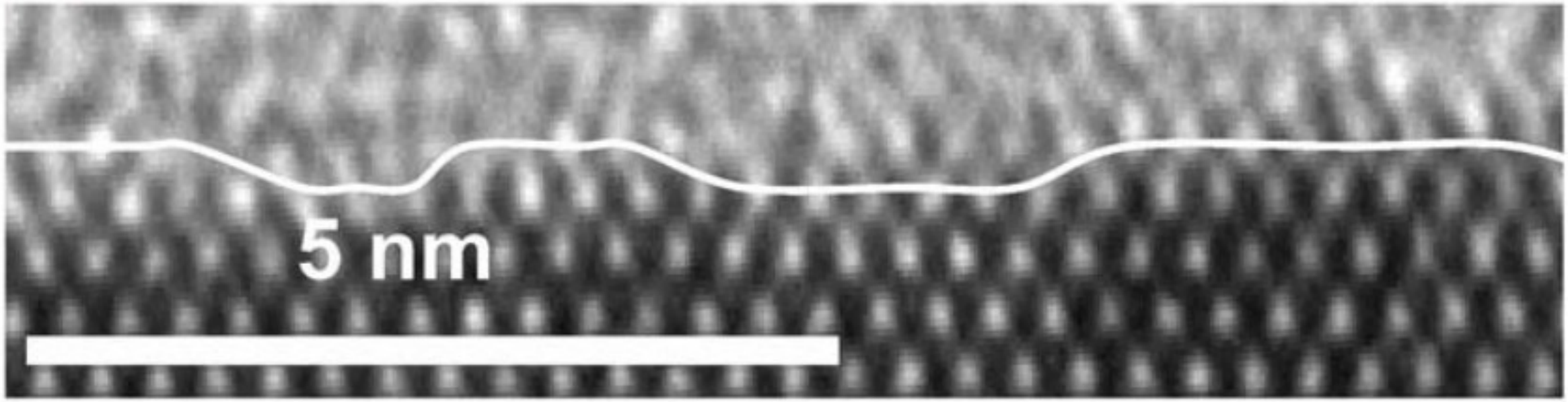
Why is the mobility position-dependent?

- In addition to the bulk scattering mechanisms,
  - Additional scattering mechanisms are important.



Rough surface

- High-resolution TEM image of SiO_2/Si interface
  - How can we characterize the roughness? Autocorrleation function, $\Delta(r)$. Usually, $\Delta^2 \exp\left(-\frac{r^2}{\Lambda^2}\right)$ or $\Delta^2 \exp\left(-\frac{r}{\Lambda}\right)$



HRTEM image (Prof. Shinichi Takagi's group, IEEE TED, 2010)

Effective mobility against effective field

- Effective field
  - Average electric field perpendicular to the Si-SiO_2 interface experienced by the carriers in the channel

$$\mathcal{E}_{eff} = \frac{1}{\epsilon_{si}}\left(|Q_d| + \frac{1}{2}|Q_i|\right)$$

Taur, Eq. (5.51)

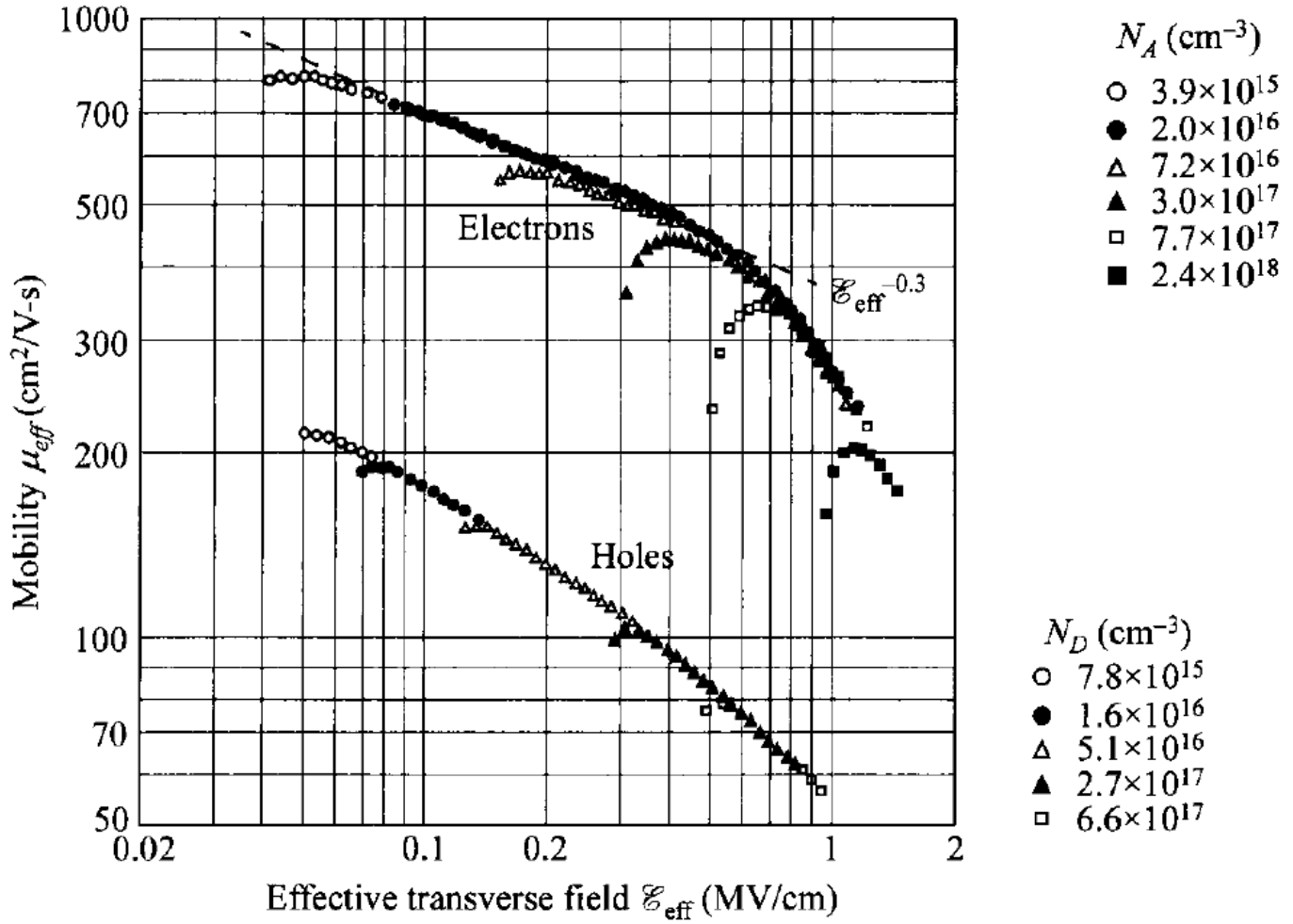
  - Using $|Q_d| \approx C_{ox}(V_t - V_{fb} - 2\phi_B)$ and $|Q_i| \approx C_{ox}(V_{gs} - V_t)$,

$$\mathcal{E}_{eff} = \frac{V_t - V_{fb} - 2\phi_B}{3t_{ox}} + \frac{V_{gs} - V_t}{6t_{ox}}$$

Taur, Eq. (5.53)

Mobility variation

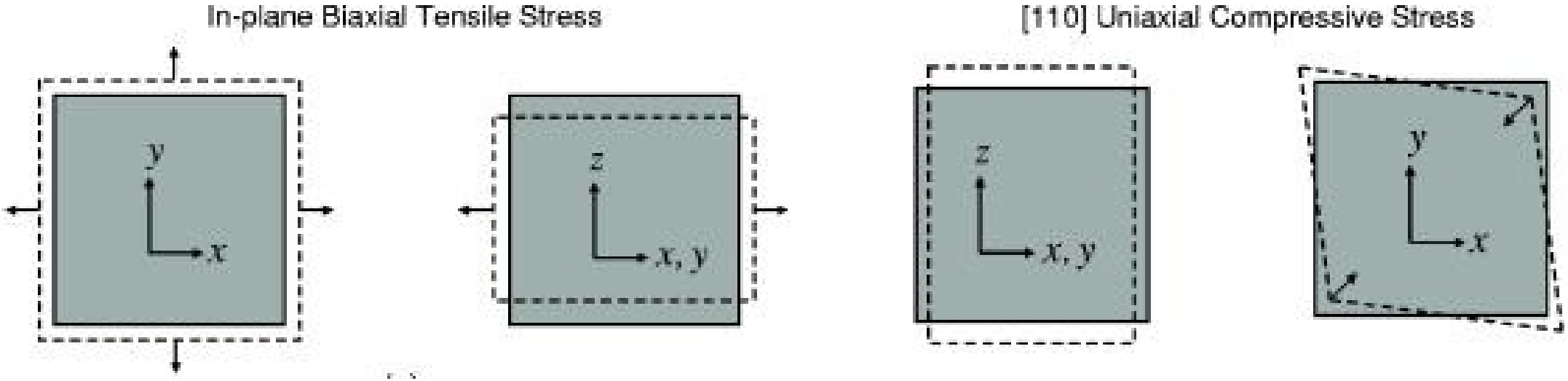
- Mobility variation (Vertical field dependence)



Inversion-layer mobility (Sze's book)

Strain effect on mobility

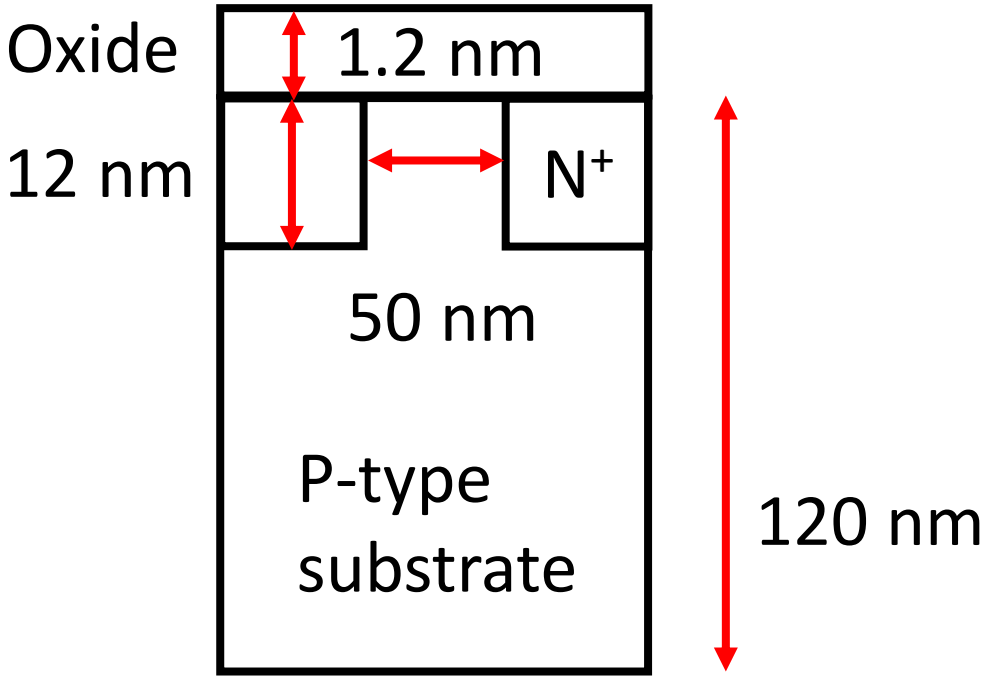
- Default wafer orientation is in the (001) plane.
  - Overall, the VLSI industry has utilized strain to gain about 10~25 % on the drive current of NMOSFET and 50 % or more on the drive current of PMOSFET.



Cubic crystals under stress (Sun et al., JAP, 2007)

Model planar MOSFET

- Effective oxide thickness of 1.2 nm
  - Gate workfunction of 4.3 eV
  - Substrate doping of $1.5X10^{18}$ cm^{-3}
  - V_{DD} of 1.2 V
- Channel length of 50 nm
  - Comparison with 1 μm device

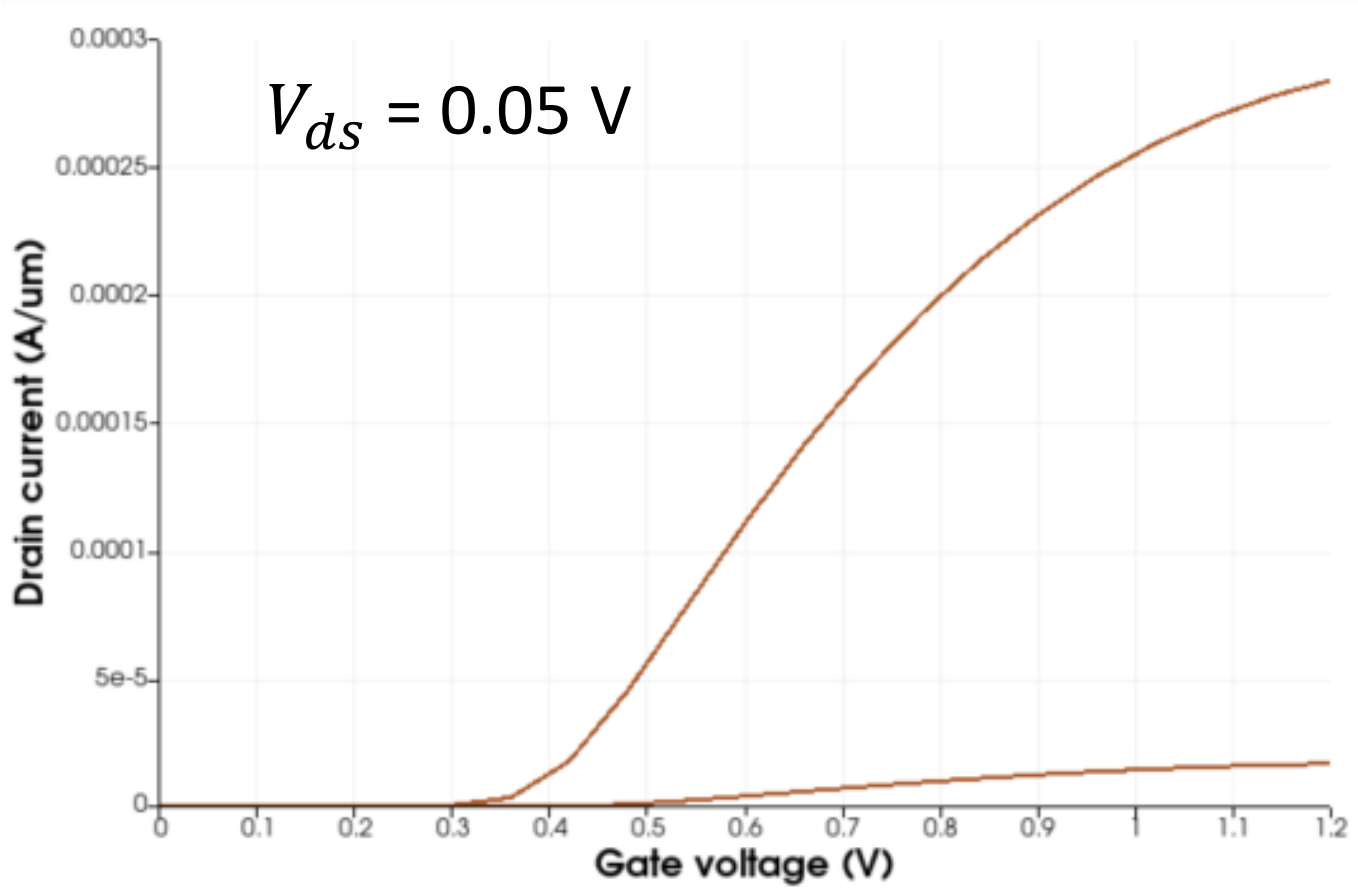


Short and long structures

- Drawn in the same scale

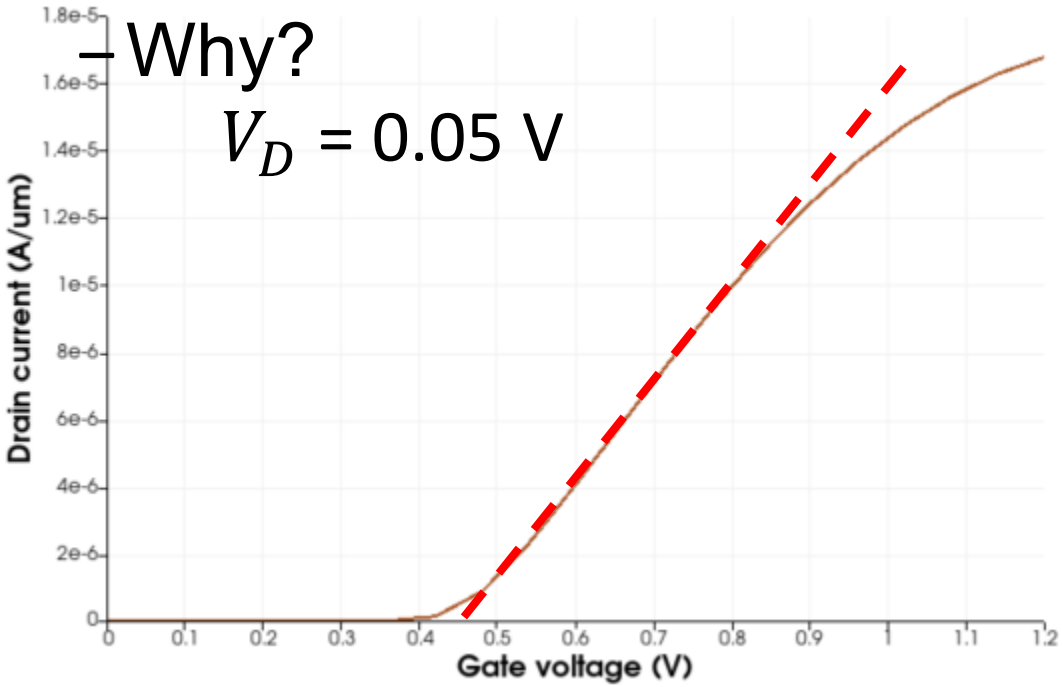
Input characteristics

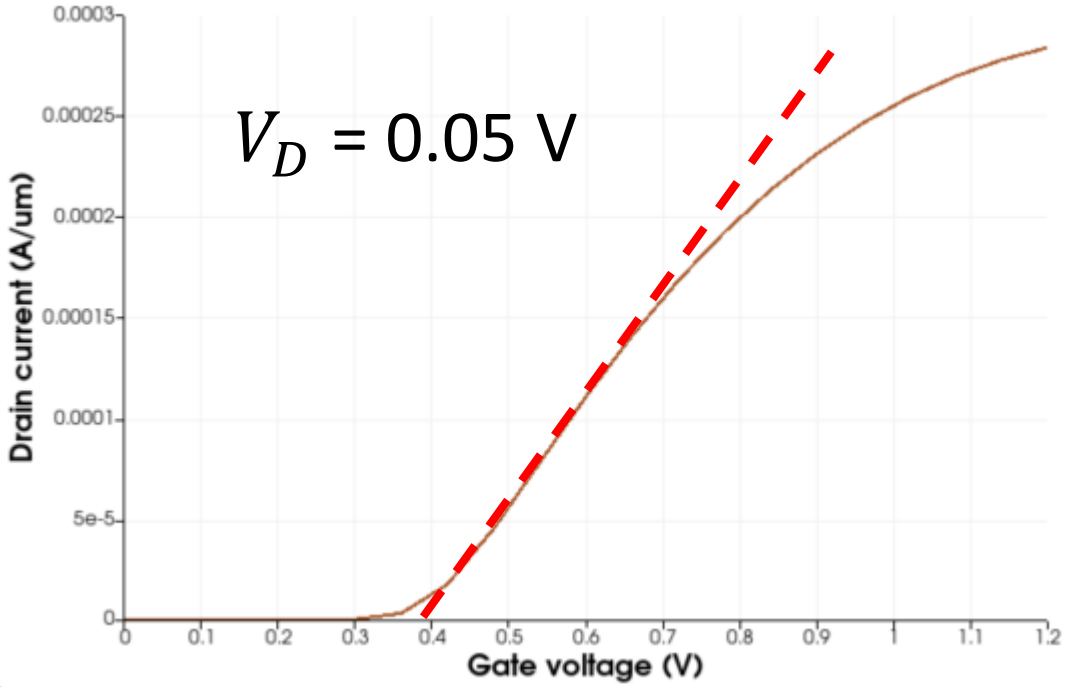
- Physical models included
  - Note that $I_D \propto \frac{1}{L}$.



Reduction of V_t, even at a low V_{ds}

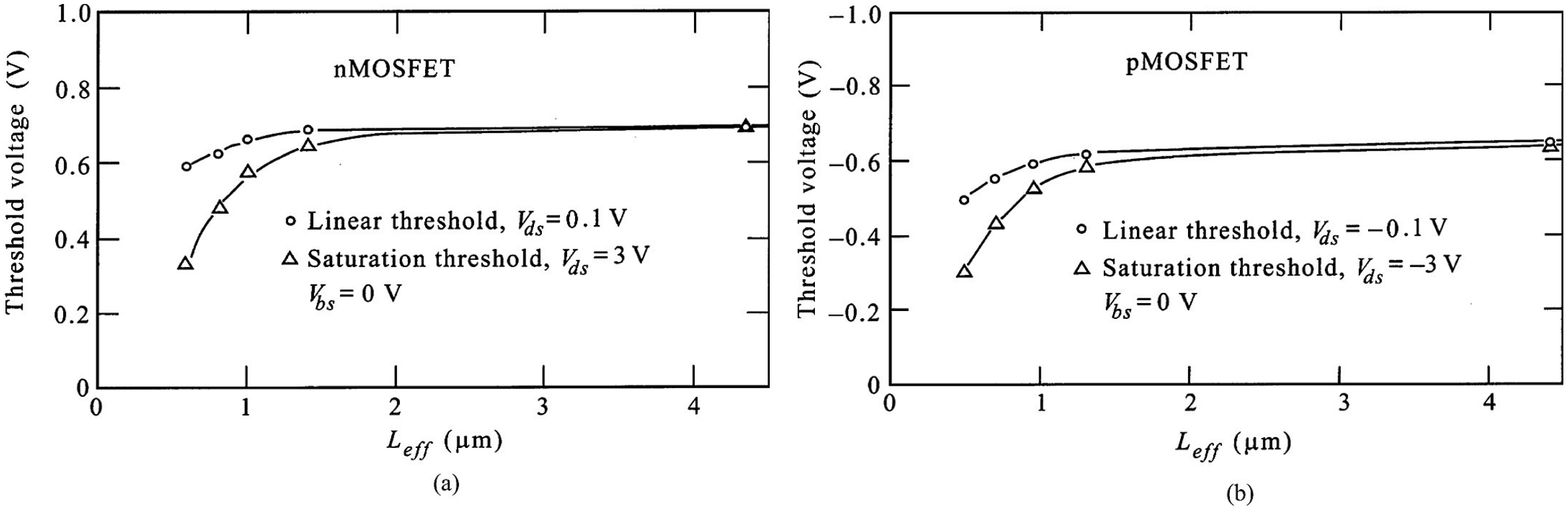
- For a short-channel device, the threshold voltage decreases.
  - ~ 0.45 V (Long-channel)
  - ~ 0.4 V (Short-channel)
  - Why?





V_t as a function of L

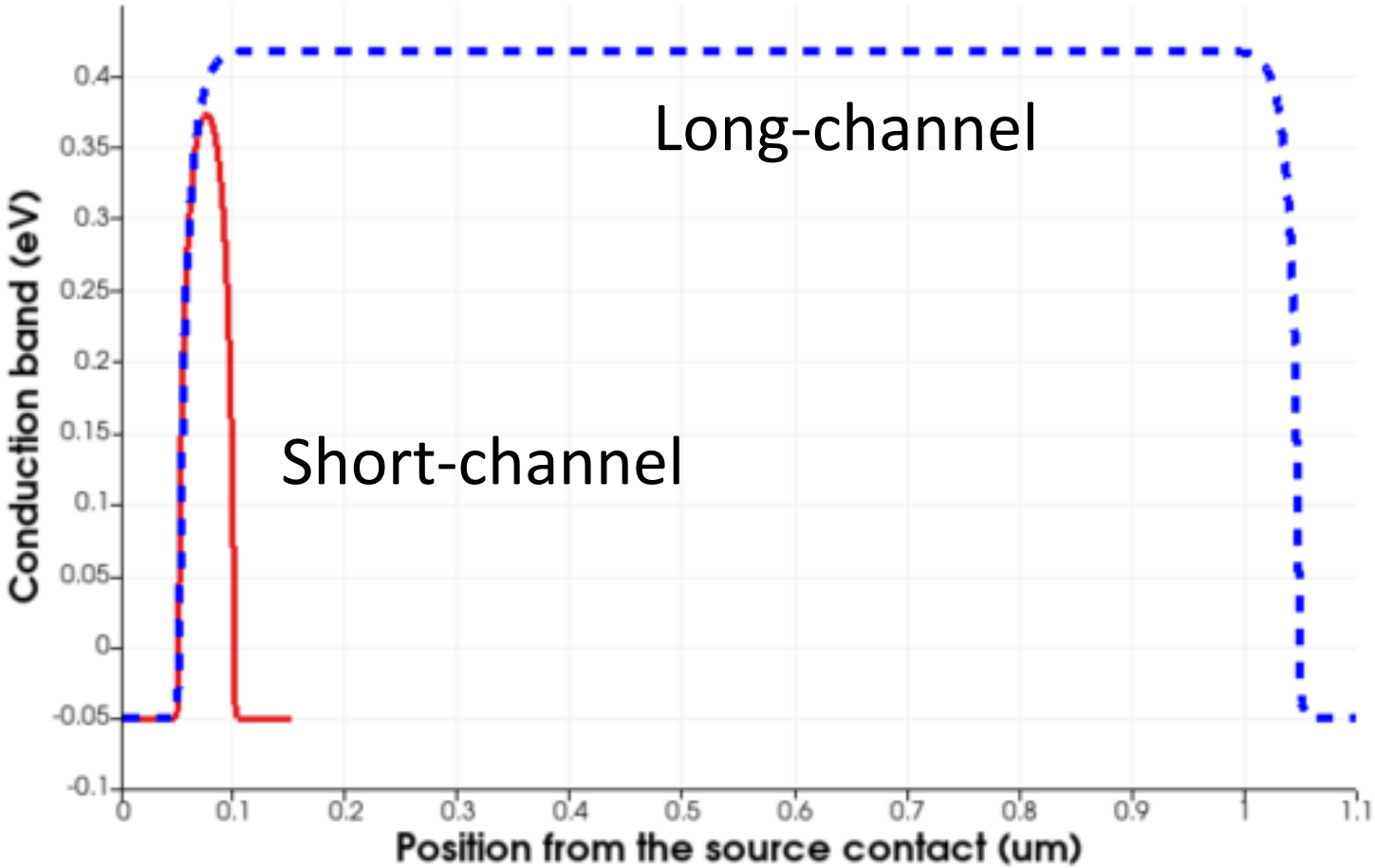
- Short-Channel Effect (SCE) = Decrease of V_t as the channel length is reduced



Short-channel threshold rolloff (Taur, Fig. 6.1)

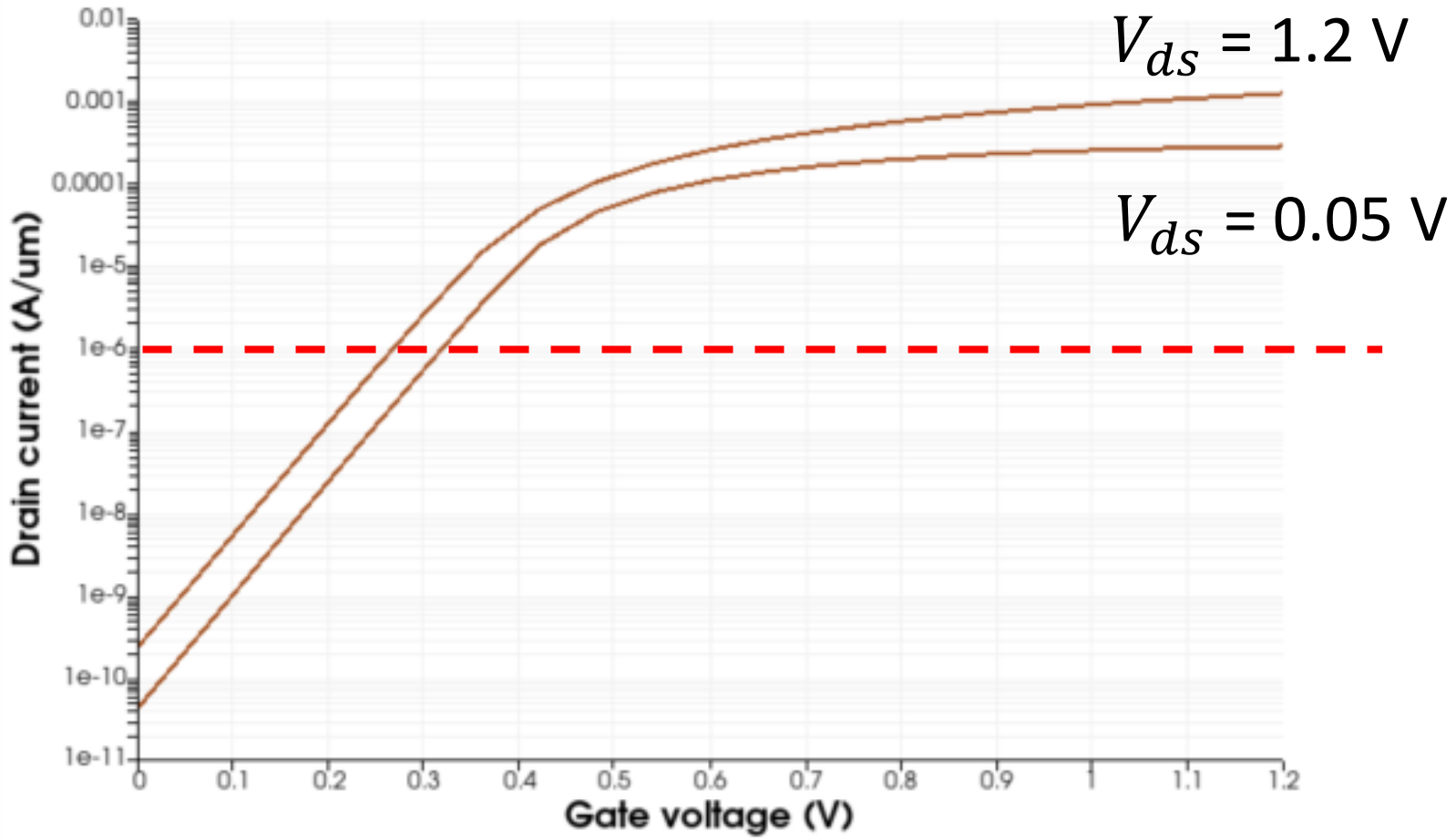
Charge sharing

- Even at V_{gs} = V_{ds} = 0 V, the conduction band profiles are different.



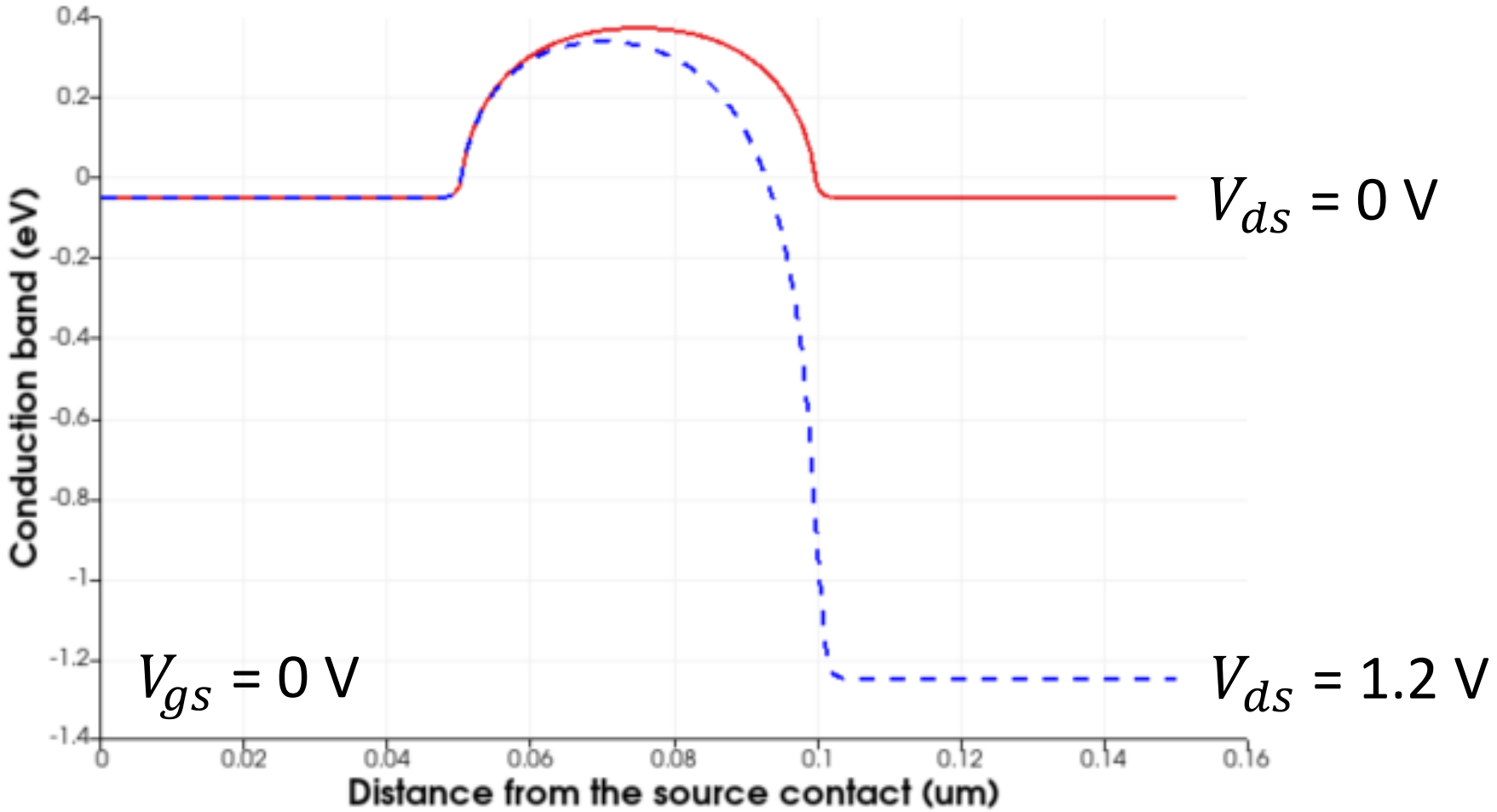
Drain-induced barrier lowering (DIBL)

- Much worse than the long-channel device
  - 45 mV/V @ I_D = 10^{-6} A/μm



Conduction band, again

- At a high V_{ds}, the energy barrier is further reduced.



Thank you!